THE HIGHER DIRECT IMAGES OF A QUASI-COHERENT MODULE ALONG AN AFFINE MORPHISM OF ALGEBRAIC SPACES VANISH IN POSITIVE DEGREES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- An affine morphism of algebraic spaces is representable. Standard, recalled at proof line 1530. It puts the statement in the representable case, where it follows from the general discussion of higher direct images for representable morphisms.
- The general representable case, and the scheme statement for relative affine vanishing. Same chapter spaces-cohomology.tex, equation equation-representable-higher-direct-image (invoked at proof line 1531) and Cohomology of Schemes, label coherent-lemma-relative-affine-vanishing (invoked at proof line 1532). Together they give the vanishing for the corresponding affine morphism of schemes, which is the assertion.

Source reference. Original lemma 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Navigation pointer for \ref{coherent-lemma-relative-affine-vanishing} used inside the counted span (line 1532); the argument that consumes it is retained in the span.

Source reference. Original equation 071Z: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 318–318. Navigation pointer for \ref{equation-representable-higher-direct-image} used inside the counted span (line 1531); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an affine morphism of algebraic spaces over S. Then $R^i f_* \mathcal{F} = 0$ for $i > 0$ and any quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$.*

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Proof. Recall that an affine morphism of algebraic spaces is representable. Hence this follows from (071Z) and Cohomology of Schemes, Lemma 01XC. □